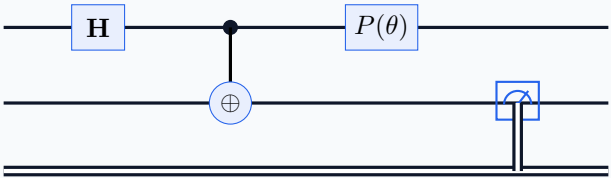


QPU Language Reference

Complete Protocol and Truth-Table Syntax



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The exact QPU protocol surface: references, constants, flags, gates, process composition, truth-table metadata, compatibility behavior, diagnostics, and quick-reference grammar.

Contents

1	Scope and execution model	1
1.1	Three levels of support	1
1.2	QPU instruction inventory	1
2	Lexical and file syntax	1
2.1	Logical lines, whitespace, and case	1
2.2	Hash comments	2
2.3	Block comments	2
2.4	PARAMS:: the process input declaration	2
2.5	Protocol file forms	2
3	Names and token references	3
3.1	Token names	3
3.2	Direct references	3
3.3	Parameter references with a dollar sign	3
3.4	Cycle-qualified references with :	4
3.5	Aliases created by SET	4
3.6	Numeric workspace references	4
3.7	Process scope and child-local references	5
4	Built-in state constants	5
4.1	0p: the zero basis state	5
4.2	1p: the one basis state	5
4.3	sp: equal superposition	5
4.4	Dimension-suffixed constants	6
4.5	Constants and state parameters	6
5	Instruction flags and argument boundaries	6
5.1	-I: input and control references	6
5.2	-O: output and target references	6
5.3	-IF: classical feed-forward (optional)	7
5.4	Where a flag list ends	7
5.5	No-parameter-substitution marker	7
5.6	Positional arguments	7
6	Gate instruction grammar and arity	7
6.1	General gate shape	7
6.2	Why arity matters	8
6.3	Validation and common mistakes	8
7	Single-qubit primitive gates	8
7.1	Pauli X (X)	8
7.2	Pauli Y (Y)	9

7.3	Pauli Z (Z)	9
7.4	Hadamard (H)	9
7.5	S phase (S)	10
7.6	T phase (T)	10
7.7	Arbitrary phase (PHASE)	11
7.8	X rotation (RX)	11
7.9	Y rotation (RY)	11
7.10	Z rotation (RZ)	12
7.11	Dagger and inverse syntax	12
8	Controlled and two-qubit primitive gates	12
8.1	Controlled NOT (CNOT)	12
8.2	Doubly controlled NOT / Toffoli (CCNOT)	13
8.3	Controlled Z (CZ)	13
8.4	Controlled phase (CPHASE)	14
8.5	Controlled Y (CY)	14
8.6	Swap (SWAP)	14
9	Derived logical gates	15
9.1	Logical NOT (NOT)	15
9.2	Reversible AND (AND)	15
9.3	Reversible NAND (NAND)	16
9.4	Reversible OR (OR)	16
9.5	Reversible XOR (XOR)	17
10	Measurement gate	17
10.1	Measurement (MEASURE)	17
11	State, token, and cycle commands	18
11.1	SET — initialize or alias (lowered/executed)	18
11.2	CREATETOKEN — reserve wires (lowered/executed)	18
11.3	DELETETOKEN — end a token name (lowered/executed)	18
11.4	FREE — same lifetime rule (lowered/executed)	18
11.5	INCREASECYCLE — cycle boundary (lowered/executed)	19
11.6	Cycle-based state evolution	19
11.7	JOIN — name an ordered register (lowered/executed)	19
11.8	SPLIT — name a subsystem (lowered/executed)	19
12	Reset: an operation that is not a gate	19
12.1	Reversible gates versus RESET	20
12.2	Where RESET comes from	20
12.3	Reset buttons compared	20
13	Process composition and return values	21
13.1	MAIN-PROCESS (lowered/executed)	21

13.2	DECLARECHILD (lowered/executed)	22
13.3	RUNCHILD — expand a child process (lowered/executed)	22
13.4	CALL — alternate child invocation (lowered/executed)	22
13.5	RETURNVALS (lowered/executed)	22
13.6	ACCEPTVALS (lowered/executed)	22
13.7	Nested full-adder example	22
13.8	Bounded child-process recursion (lowered/executed)	23
13.8.1	Recursion on the circuit canvas	24
13.9	Structured classical branches: IF / ELSE / ENDIF (lowered/executed)	25
13.9.1	IF / ELSE on the circuit canvas	25
13.10	Worked example: recursion, measurement, and IF / ELSE together	25
14	Custom gates: packaging a circuit as one block	26
14.1	Register one in four steps	26
14.2	How the process maps onto wires	27
14.3	Reasoning about reversibility	27
14.4	Rules to remember	28
15	Process control and checkpoints	28
15.1	COMPILEPROCESS (lowered/executed)	28
15.2	MASTerval (lowered/executed)	28
15.3	Save-state checkpoint instruction (lowered/executed)	28
15.4	Load-state checkpoint instruction (lowered/executed)	28
16	Truth-table companion syntax	29
16.1	Text .qpuio structure	29
16.2	MAIN-PROCESS: table header	29
16.3	INPUTS: and OUTPUTS: column groups	29
16.4	Column header row	29
16.5	Indexed data rows	29
16.6	CSV spelling	30
16.7	Partial truth tables	30
16.8	JSON .qpuio envelope	30
16.9	How a table test runs	31
16.10	Protected bundled tables	31
17	Compiler output and lowering details	31
18	Error conditions and practical rules	31
19	Quick-reference grammar	32
20	Glossary	32
21	Operation checklist	34

1 Scope and execution model

The compiler turns a plain-text `.qpucir` protocol into one flat sequence of circuit gates:

protocol source $\longrightarrow$ logical lines $\longrightarrow$ QPU instructions $\longrightarrow$ scoped tokens $\longrightarrow$ gates $\longrightarrow$ state-vector simulation.

Child processes are expanded at compile time. The UI and simulator therefore see the same ordered gate list even when the source contains nested processes.

1.1 Three levels of support

Status	Meaning
lowered/executed	The QPU language accepts the instruction and the compiler emits behavior that the simulator executes.
accepted, not executed	The QPU language accepts the instruction and preserves it in compiler output or the compiler log, but no visual/simulator operation is emitted yet.
compiler-internal	The compiler may emit the operation itself, but protocol source cannot name it directly. RESET is the important example.

1.2 QPU instruction inventory

Command	Status	Purpose
X, Y, Z, H, S, T, PHASE	lowered/executed	Single-qubit matrix operations.
CNOT, CCNOT, CZ, CY, SWAP	lowered/executed	Controlled and two-wire primitive operations.
NOT, AND, NAND, OR, XOR	lowered/executed	Derived logical target mutations.
MEASURE	lowered/executed	Measure one named wire or all known wires.
SET, CREATETOKEN	lowered/executed	Initialize/alias tokens and reserve wires.
RUNCHILD, CALL	lowered/executed	Expand a catalog/library child process.
RETURNVALS, ACCEPTVALS	lowered/executed	Bind child and process outputs.
MAIN-PROCESS, INCREASECYCLE	lowered/executed	Mark a process and flush cycle workspace.
DECLARECHILD	lowered/executed	Bind a catalog child for later RUNCHILD/CALL.
FREE, DELETETOKEN	lowered/executed	End a name. Reuse its wire only from a known $ 0\rangle$.
JOIN, SPLIT	lowered/executed	Name an ordered register and a dimension-sized part of it.
COMPILEPROCESS	lowered/executed	Verify a child process without inlining its gates.
MASTerval	lowered/executed	Distinguished return. Does not add a truth-table column.
SAVE_STATE, LOAD_STATE	lowered/executed	Simulator checkpoints of the state vector and measurements.

Not source syntax: RESET exists in the gate registry, but it is not a supported source instruction. Write `SET wire 0p`; the compiler schedules internal RESET gates for zeroed workspace.

2 Lexical and file syntax

2.1 Logical lines, whitespace, and case

Commands and flags are case-insensitive and are separated by whitespace. Token names retain their spelling. CRLF and LF line endings are accepted. A physical line ending *exactly* in a backslash joins to the next line:

```
PARAMS: A:state B:state \
        Cin:state
CCNOT -I A B \
      -O Carry
```

The continuation backslash must be the final character; spaces after it prevent continuation. Blank logical lines are discarded.

2.2 Hash comments

```
# Whole-line or trailing hash comment
H -I Q:0 -O Q:0 # prepare |+>
```

A hash starts a comment that continues through the logical line. Comments are for explanations and do not become instructions. Because continuation joining happens first, a hash on a continued logical line can hide text that appeared on a later physical line. Prefer comments after a complete instruction.

2.3 Block comments

```
/* A block comment may span logical lines. */
PHASE=pi/2 -I Q:0 -O Q:0
```

Text from `/*` through `*/` is ignored. A block may occupy one line or continue across several lines. Block comments are useful for temporarily annotating a region, but they should not be nested. Keep delimiters visually obvious so an omitted closing marker does not hide the remainder of a file.

2.4 PARAMS:: the process input declaration

```
PARAMS: A:state B:state Theta:float Count:int
```

`PARAMS:` is recognized only when it is the first non-comment logical line. Each whitespace-separated entry containing a colon becomes `name:type`. Declaration order is interface order: `RUNCHILD` inputs and truth-table input columns map from left to right.

Long declarations can use continuation:

```
PARAMS: A0:state A1:state B0:state B1:state \
        Cin:state Width:int
```

The system retains the type text:

- `state` entries become user-facing circuit inputs and truth-table input columns. Each participating state parameter resolves to a qubit wire.
- `int` and `float` entries are retained but excluded from user-facing qubit controls and truth-table state inputs. They describe numeric process metadata rather than prepared qubits.
- other type names are retained as written. Because they are not one of the recognized numeric exclusions, they may be exposed like stateful parameters; prefer `state`, `int`, or `float` for predictable behavior.

`PARAMS` has a colon after the keyword; `MAIN-PROCESS` does not. An empty or misplaced `PARAMS` line is not a substitute for a valid first-line declaration. The process's `MAIN-PROCESS` instruction is described in the composition chapter.

2.5 Protocol file forms

The normal authoring format is plain UTF-8 QPU text saved as `ProcessName.qpucir`. A `.qpucir.txt` download is also plain source and is useful where browsers or editors insist on a text extension. Uploading plain text derives the display name from `MAIN-PROCESS`.

The loader can also accept a JSON export envelope:

```
{
  "format": "qpucir",
  "version": 1,
  "name": "First circuit",
  "source": "MAIN-PROCESS FirstCircuit\nRETURNVALS Q"
}
```

The source string is the authoritative language text. Export envelopes may also contain compiled qubit count, gates, token map, and export timestamp metadata, but compilation from source determines current behavior.

3 Names and token references

A *token* is the QPU language's name for a logical storage location. For the circuits in this guide, a token normally identifies one qubit wire. Instructions refer to tokens instead of physical machine addresses. During compilation, the system resolves each live token to an integer qubit index. This indirection makes protocols readable and lets child processes have private workspace without colliding with their parents.

3.1 Token names

A bare token is written as a whitespace-delimited name:

```
Q
Control
CarryOut
scratch_2
0
```

Names are case-sensitive even though instruction names are not: **Carry** and **carry** are distinct tokens. A name cannot contain whitespace because whitespace separates language elements. Avoid names that begin with a hyphen: inside a flag list, any hyphen-prefixed token is treated as the beginning of another flag.

Tokens are allocated lazily. Mentioning a token in a gate, a state-setting instruction, or **CREATETOKEN** gives it a wire when compilation needs one. Consequently, source order affects the internal numeric wire index, but not the meaning of a consistently named circuit.

3.2 Direct references

A direct reference is simply the token's name:

```
SET Q Op
H -I Q -O Q
MEASURE -I Q
RETURNVALS Q
```

Every occurrence resolves to the same token within that process invocation. For a one-qubit gate, repeat the name after both **-I** and **-O**. This does not copy the state. It says that **Q** is the instruction's incoming wire and that the same **Q** wire is mutated in place.

3.3 Parameter references with a dollar sign

A leading dollar sign is the explicit parameter/reference spelling:

```
PARAMS: Input:state
MAIN-PROCESS DollarReference
SET Local $Input
H -I $Input -O $Input
RETURNVALS Local
```

Address resolution removes the leading \$. Thus `Input` and `$Input` ultimately refer to the same declared parameter. The dollar form is especially useful inside reusable child protocols because it visually separates caller-supplied values from child-local workspace.

`SET Local \ $Input` creates an alias: `Local` resolves to the same underlying wire as `Input`. It does *not* clone an unknown quantum state. Quantum states cannot generally be copied because of the no-cloning theorem; aliasing avoids claiming that a physical copy occurred.

3.4 Cycle-qualified references with :

A token may carry a suffix after a colon:

```
Q:0
Q:1
Carry:next
```

The current system strips the colon and everything after it when mapping source references to visual qubit wires. Therefore all three examples above whose base is `Q`—`Q:0`, `Q:1`, and bare `Q`—identify the same logical wire. The suffix is protocol notation for a stage, cycle, or bit version; it does not create a second qubit.

```
SET Q:0 Op
H -I Q:0 -O Q:1
MEASURE -I Q:2
```

This example applies `H` to one wire and then measures that same wire. The changing suffix communicates progression to a human reader, while `INCREASECYCLE` provides the actual compiler cycle boundary.

Practical rule: if two values must exist simultaneously, give them different base names. Do not expect `Q:0` and `Q:1` to hold two independent states.

3.5 Aliases created by SET

When the second operand of `SET` is not a built-in state constant, `SET` creates a name alias:

```
SET Working Input
SET Output Working
X -I Output -O Output
```

After resolution, `Working`, `Output`, and `Input` name one wire. Mutating any alias mutates that wire for every alias. This behavior is useful for readable names and process binding, but it is not assignment in the classical “copy a value into a new variable” sense.

Aliases are resolved within a process frame. `PARAMS` and parent/child bindings take precedence over ordinary child-local scoping, ensuring that an operation on a bound parameter affects the caller’s intended wire.

3.6 Numeric workspace references

Bare numeric names such as `0`, `1`, and `2` are valid token names and are common in reusable arithmetic children:

```
SET 0:0 $A
SET 1:0 $B
SET 2:0 $Cin
```

When a child process uses a numeric workspace name, the compiler places it in the parent invocation’s workspace namespace. This lets nested adders reuse the same compact source names without globally identifying every child invocation’s scratch wires.

Numeric token names are not number literals. `2` names a wire; it does not mean the integer two or the basis state $|2\rangle$.

3.7 Process scope and child-local references

Each process invocation receives a unique scope. A local name such as `Scratch` in two separate child calls therefore denotes two separately resolved local references unless output or parameter binding deliberately joins them. Conceptually, compilation turns:

```
RUNCHILD Worker -I A -O First
RUNCHILD Worker -I B -O Second
```

into two independent Worker frames. This is similar to local variables in two function calls. PARAMS map caller references into the frame, and RETURNVALS map selected frame references back out.

4 Built-in state constants

State constants describe how a wire begins. They are not ordinary token names, even though they participate in address resolution. Constants are case-insensitive and may be written with a leading dollar sign. The three supported base constants are `0p`, `1p`, and `sp`; the letter `p` distinguishes protocol state notation from ordinary numbers.

4.1 0p: the zero basis state

`0p` denotes

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

```
SET Output 0p
```

For workspace, this schedules zero preparation. The compiler batches pending zero preparations into internal RESET operations immediately before the next real gate, at an `INCREASECYCLE`, or at process end. RESET is deliberately not source syntax; `SET Output 0p` is the public way to request zero.

Zero is the usual target initialization for reversible AND, OR, and XOR: with $t = 0$, the mutation $t \leftarrow t \oplus f(x)$ leaves $t = f(x)$.

4.2 1p: the one basis state

`1p` denotes

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

```
SET Flag 1p
```

The compiler realizes one preparation by applying X to a zero-initialized wire. A target beginning in one is useful when the desired reversible result is the complement of a Boolean predicate. For example, $1 \oplus (A \wedge B) = \neg(A \wedge B)$, which is NAND.

4.3 sp: equal superposition

`sp` denotes the plus state

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

```
SET Probe sp
```

The compiler realizes this preparation with a Hadamard gate. If immediately measured, the result is 0 or 1 with equal probability. It is not merely a hidden classical random bit: both amplitudes coexist and can interfere after later phase and Hadamard operations.

In an expected truth-table *output* cell, **sp** has a different testing role: it means either measured bit is acceptable. Context therefore matters—protocol SET initializes a plus state, while expected output metadata uses **sp** as a don’t-care marker.

4.4 Dimension-suffixed constants

The language also recognizes:

```
Op_dim2
1p_dim8
sp_dim4
```

The suffix must be `_dim` followed by a power of two that is at least 2. That number is a Hilbert-space dimension, so `_dim4` prepares two qubits and `_dim8` prepares three. `Op` prepares every wire in $|0\rangle$, `1p` flips only the least-significant wire, and `sp` applies H to every wire. `_dim2` is the same one-wire preparation as the unsuffixed constant. Any other dimension is rejected. Wire k of register **R** is named `R[k]`.

4.5 Constants and state parameters

For an ordinary local token, SET emits preparation behavior. For a PARAMS entry typed **state**, SET supplies a default instead:

```
PARAMS: Input:state
MAIN-PROCESS DefaultInput
SET Input Op
X -I Input -O Input
RETURNVALS Input
```

The UI or truth-table runner may replace the default start state for Input. This distinction is what allows one compiled protocol to be tested over every binary input row instead of hard-wiring its parameter to zero.

5 Instruction flags and argument boundaries

5.1 -I: input and control references

`-I` begins the instruction’s input list:

```
H      -I Target      -O Target
CNOT -I Control      -O Target
CCNOT -I A B          -O Target
```

Its interpretation depends on the gate. For a one-qubit gate, the input names the same wire that the output mutates. For CNOT/CZ/CY it names one control. For CCNOT and the two-input derived logic gates it names two controls. Input order does not matter to symmetric predicates such as AND or XOR, but it remains part of the written instruction and should be kept consistent for readability.

5.2 -O: output and target references

`-O` begins the output list. For most gates its first token is the target wire whose amplitudes are changed:

```
X -I Q -O Q
AND -I A B -O Result
```

`-O` does not mean “allocate a fresh classical output.” The target may already contain a quantum state, and reversible logic XORs its result into that existing state. SWAP is exceptional: it requires two outputs matching its two inputs. MEASURE takes no output flag because the measurement map stores its classical result.

5.3 -IF: classical feed-forward (optional)

A gate may carry a forward-only classical condition after a prior MEASURE:

```
X -I Bob -O Bob -IF Alice=1
Z -I Bob -O Bob -IF M0=1
```

The named token must already have been measured; running a conditioned gate before its MEASURE is a runtime error. If the classical bit does not match, the gate is skipped and execution continues. The skipped gate stays in the circuit. To condition several gates, or to pair a gate with an alternative, use the structured IF / ELSE / ENDIF block described in Structured classical branches; it lowers to the same -IF conditions. Either way there is no backward edge—the circuit remains a finite ordered list.

5.4 Where a flag list ends

An -I or -O list continues until the next token beginning with a hyphen or the end of the line:

```
CCNOT -I A B -O Carry -$R
```

Here A and B belong to -I, Carry belongs to -O, and -\$R starts a separate flag. This rule is why token names should not begin with a hyphen.

Flags are case-insensitive, so -i and -o work, but uppercase is the documented style. Supplying flags in an unusual order may still parse, but -I followed by -O is the clearest and matches generated source.

5.5 No-parameter-substitution marker

```
PHASE=pi/4 -I Q -O Q -$R
```

The language recognizes this marker and, on that instruction only, resolves names as locals instead of substituting PARAMS bindings. Aliases created by SET still apply.

5.6 Positional arguments

Not every instruction uses flags. SET, MAIN-PROCESS, RETURNVALS, ACCEPTVALS, and several compatibility instructions consume positional words:

```
SET Target Op
MAIN-PROCESS Example
RETURNVALS Carry Sum
```

The command name is the first word and positional arguments follow in order. RETURNVALS order is part of the process interface. For non-gate compatibility instructions, extra words may be retained even when no execution behavior is defined; accepted text should not be mistaken for an implemented effect.

6 Gate instruction grammar and arity

6.1 General gate shape

```
GATE -I input_or_controls -O target
```

The uppercase word selects a gate. -I lists the incoming wire for a one-qubit operation or the controls for a controlled operation. -O names the wire that is mutated. Except for SWAP, gate instructions do not create a separate output wire automatically.

```
H -I Q -O Q
CNOT -I Control -O Target
```

In the first row, Q is listed twice because the state flows through H on one wire. In the second, Control is preserved and Target is conditionally changed.

6.2 Why arity matters

Gate family	Required -I count	Required -O count
X, Y, Z, H, S, T, PHASE, NOT	1	1
CNOT, CZ, CY	1	1
CCNOT, AND, NAND, OR, XOR	2	1
SWAP	2	2
MEASURE	0 or 1	0

All counts are exact except **MEASURE**. For ordinary single-qubit gates, name the same wire in -I and -O. The output is the mutated target; for controlled gates, -I contains controls.

6.3 Validation and common mistakes

Arity is part of the language contract. A CNOT with two inputs is not silently converted into CCNOT, and a CCNOT with one input is not converted into CNOT. The compiler reports whether an instruction has too few or too many inputs or outputs.

```
# Wrong: H needs one output.
H -I Q

# Wrong: CNOT needs exactly one control.
CNOT -I A B -O Target

# Right:
CCNOT -I A B -O Target
```

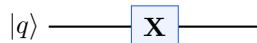
The target is removed from the compiler's control list if the same resolved wire appears on both sides. This is necessary for the normal one-qubit form. For controlled gates, do not also list the target as an input; doing so either violates arity or obscures the intended control relationship.

7 Single-qubit primitive gates

Placing a single-qubit gate

X, Y, Z, H, S, T, RX, RY, RZ, and PHASE act on one wire, so the workbench uses only **Target particle**; **Control A** and **Control B** are ignored. PHASE, RX, RY, RZ, and CPHASE also show a **Phase angle** slider. On the canvas, drop the block on the wire you want to change.

7.1 Pauli X (x)



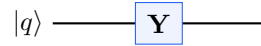
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle.$$

X is the quantum bit flip. On the Bloch sphere it is a half-turn around the x-axis. It exchanges the north and south poles and is self-inverse: $X^2 = I$. Unlike a classical NOT implementation, X remains meaningful on a superposition because it exchanges amplitudes linearly without measuring.

```
X -I Q:0 -0 Q:0
```

Input	Classical comparison	Output
0	NOT 0	1
1	NOT 1	0

7.2 Pauli Y (Y)



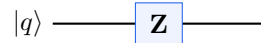
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle.$$

It flips the basis value like X but adds phases that become observable through later interference. Geometrically it is a half-turn around the y-axis and is self-inverse, $Y^2 = I$. The factors i and $-i$ are global phases for either basis state by itself, but on components of a larger superposition they may become relative phases.

```
Y -I Q -0 Q
```

Input	Measured bit	Quantum output
0	1	$i 1\rangle$
1	0	$-i 0\rangle$

7.3 Pauli Z (Z)



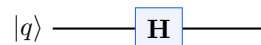
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Z does not change a computational-basis measurement, but it changes relative phase. For example, $Z|+\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. On the Bloch sphere it is a half-turn around z. It is self-inverse and is the simplest example showing why quantum state cannot be reduced to a table of immediate measurement probabilities.

```
Z -I Q -0 Q
```

Input	Classical bit	Quantum output
0	0	$ 0\rangle$
1	1	$- 1\rangle$

7.4 Hadamard (H)



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle.$$

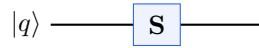
Hadamard changes between the computational z basis and the plus/minus x basis. It creates equal magnitudes from a basis state, but the sign of the second amplitude depends on the input. It is also self-inverse: applying H twice returns the original state. H is commonly used before controlled gates to create entanglement and after phase gates to reveal interference.

```
SET Q 0p
H -I Q -0 Q
MEASURE -I Q
```

Input	Measurement distribution	State before measurement
0	50% 0, 50% 1	$(0\rangle + 1\rangle)/\sqrt{2}$
1	50% 0, 50% 1	$(0\rangle - 1\rangle)/\sqrt{2}$

Equal classical probabilities hide the relative minus sign responsible for interference.

7.5 S phase (S)



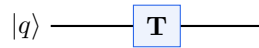
$$S = P(\pi/2) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad S|1\rangle = i|1\rangle, \quad S^2 = Z.$$

S is a quarter-turn around the Bloch sphere's z-axis. It changes $|+\rangle$ into $(|0\rangle + i|1\rangle)/\sqrt{2}$ while leaving basis measurement probabilities unchanged. Its inverse is $S^\dagger = P(-\pi/2)$, not S itself.

```
S -I Q -0 Q
```

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$i 1\rangle$

7.6 T phase (T)



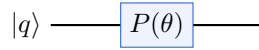
$$T = P(\pi/4) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}, \quad T^2 = S, \quad T^4 = Z.$$

T is an eighth-turn around z. Together with gates such as H and CNOT, it is an important non-Clifford resource for expressing general quantum computations. Its inverse is $T^\dagger = P(-\pi/4)$. `Tdg` and `Tinv` lower to that gate, and `Sdg` and `Sinv` lower to $S^\dagger = P(-\pi/2)$. `PHASEdg=pi/4` is the same negative phase.

```
T -I Q -0 Q
```

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$e^{i\pi/4} 1\rangle$

7.7 Arbitrary phase (PHASE)



$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}, \quad \alpha|0\rangle + \beta|1\rangle \mapsto \alpha|0\rangle + \beta e^{i\theta}|1\rangle.$$

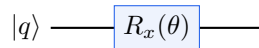
PHASE is the general form behind S, T, and Z. It preserves $|\alpha|^2$ and $|\beta|^2$ because $|e^{i\theta}| = 1$, yet rotates the relative phase by θ . On the Bloch sphere this is a z-axis rotation up to global phase. A later H or another basis-changing gate is normally needed to turn that phase difference into changed computational-basis probabilities. The angle is embedded in the opcode:

```
PHASE=1.5708 -I Q -0 Q      # radians
PHASE=3/2 -I Q -0 Q        # rational radians
PHASE=90d -I Q -0 Q        # degrees
PHASE=45/2d -I Q -0 Q      # rational degrees
PHASE=pi -I Q -0 Q
PHASE=-11pi/6 -I Q -0 Q
PHASE=2*pi/3 -I Q -0 Q
PHASEdg=pi/4 -I Q -0 Q     # P(-pi/4)
invPHASE=pi/2 -I Q -0 Q    # P(-pi/2)
```

Accepted numeric atoms include signed integers, decimals, leading-dot decimals, and one optional rational denominator. Pi notation accepts an optional sign, an integer coefficient, an optional *, and an optional positive integer denominator. A zero denominator and malformed angle raise an error.

Input	Measured bit	Quantum output
0	0	$ 0\rangle$
1	1	$e^{i\theta} 1\rangle$

7.8 X rotation (RX)

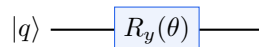


$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

RX rotates about the Bloch x -axis. RX=pi is X up to global phase.

```
RX=pi/2 -I Q -0 Q
RX=90d -I Q -0 Q
RXdg=pi/4 -I Q -0 Q
```

7.9 Y rotation (RY)

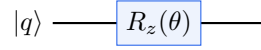


$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

RY rotates about y and is useful for preparing real superpositions from $|0\rangle$.

```
RY=pi/2 -I Q -0 Q
```

7.10 Z rotation (RZ)



$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}.$$

RZ rotates about z . It differs from PHASE by a global phase that does not affect single-qubit measurement probabilities.

```
RZ=pi/4 -I Q -0 Q
```

7.11 Dagger and inverse syntax

A reversible gate accepts **dg** (dagger) or **inv** (inverse), written after the opcode or before it. Both spellings set **reverse=true**. Self-inverse gates such as X, CNOT, AND, and OR still accept the marker so intent is visible, even though the matrix is unchanged.

```
Xdg -I Q -0 Q
invH -I Q -0 Q
CNOTdg -I A -0 T
ANDdg -I A B -0 T
dgSWAP -I A B -0 A B
Sdg -I Q -0 Q           # P(-pi/2)
Tinv -I Q -0 Q          # P(-pi/4)
PHASEdg=pi/4 -I Q -0 Q  # P(-pi/4)
```

Sdg, Sinv, dgS, and invS synthesize $S^\dagger$. Tdg, Tinv, dgT, and invT synthesize $T^\dagger$. PHASEdg=angle, PHASEinv=angle, dgPHASE=angle, and invPHASE=angle negate the angle. For these self-inverse gates the marker emits the ordinary gate, which is already its own inverse:

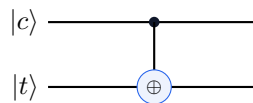
$X, Y, Z, H, \text{CNOT}, \text{CCNOT}, \text{CZ}, \text{CY}, \text{SWAP}.$

On the canvas the dagger prints $\dagger$ and uses a blue outline. While that step is active the outline is purple. A forward gate uses a red outline only while its step is active. The palette **Inverse** toggle drops the dagger instead of the forward gate.

A **dg** or **inv** marker on a reversible derived gate such as ANDdg is accepted and is self-inverse. MEASUREdg parses and warns, because measurement has no inverse. Compiler-internal RESET remains unavailable in source.

8 Controlled and two-qubit primitive gates

8.1 Controlled NOT (CNOT)



$$|c, t\rangle \mapsto |c, t \oplus c\rangle.$$

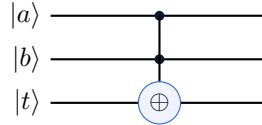
CNOT leaves the control unchanged. When the control is zero, the target is unchanged; when the control is one, X acts on the target. Because the map is a permutation of basis states, it is reversible and self-inverse. A superposition control can entangle the wires, so “classical XOR” describes only its basis-state action.

```
CNOT -I Control -0 Target
```

c	t	c'	t'
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

CNOT is a reversible embedding of XOR because it preserves the control. In the workbench, the control comes from **Control A** and the target from **Target particle**; **Control B** is ignored.

8.2 Doubly controlled NOT / Toffoli (CCNOT)



$$|a, b, t\rangle \mapsto |a, b, t \oplus (a \wedge b)\rangle.$$

CCNOT, also called Toffoli, applies X only when both controls are one. It is a reversible universal building block for classical computation embedded in a quantum circuit. The target's previous value matters: zero computes AND, while one computes NAND. The gate is self-inverse.

CCNOT -I A B -0 Target

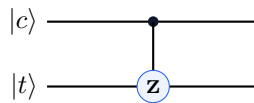
a	b	t	a'	b'	t'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

With a target initialized to zero, Toffoli computes classical AND reversibly.

Controls are preserved, the target is the output

In the table, the a' and b' columns always equal a and b : the two controls are only read. Only t can change, and it flips exactly in the two rows where $a = b = 1$. In the **Interactive workbench**, choose **CCNOT** in **Gate**, the two inputs in **Control A** and **Control B**, and the output wire in **Target particle**. The *CCNOT demo* starter circuit is exactly this gate with both controls prepared as 1.

8.3 Controlled Z (CZ)



$$\text{CZ}|c, t\rangle = (-1)^{ct}|c, t\rangle, \quad \text{CZ} = \text{diag}(1, 1, 1, -1).$$

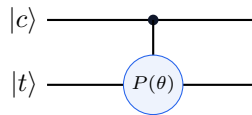
CZ adds a minus sign only to the $|11\rangle$ amplitude. No computational-basis bit changes, but superpositions can acquire observable relative phase. CZ is symmetric between its two wires, even though QPU syntax writes one as the input control and one as the output target. It is self-inverse.

CZ -I Control -0 Target

Input	Measured output	Amplitude multiplier	State
00	00	+1	$ 00\rangle$
01	01	+1	$ 01\rangle$
10	10	+1	$ 10\rangle$
11	11	-1	$- 11\rangle$

The classical columns are unchanged; the quantum phase is the operation.

8.4 Controlled phase (CPHASE)

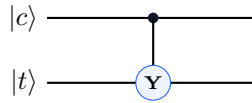


$$|11\rangle \mapsto e^{i\theta}|11\rangle,$$

while $|00\rangle$, $|01\rangle$, and $|10\rangle$ are unchanged. CZ is the special case $\theta = \pi$. CPHASE unlocks compact Quantum Fourier Transform and phase-estimation demos.

CPHASE=pi/2 -I Control -0 Target
CPHASE=90d -I Control -0 Target

8.5 Controlled Y (CY)



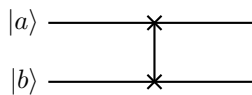
$$|0, t\rangle \mapsto |0, t\rangle, \quad |1, 0\rangle \mapsto i|1, 1\rangle, \quad |1, 1\rangle \mapsto -i|1, 0\rangle.$$

CY applies the Pauli Y rotation to the target only when the control is one. Its basis-bit pattern resembles CNOT, but the $\pm i$ factors distinguish their behavior on superpositions. Since $Y^2 = I$, CY is self-inverse.

CY -I Control -0 Target

c	t	c'	t'	Quantum phase
0	0	0	0	+1
0	1	0	1	+1
1	0	1	1	i
1	1	1	0	$-i$

8.6 Swap (SWAP)



$$\text{SWAP}|a, b\rangle = |b, a\rangle.$$

SWAP exchanges the complete quantum states of two wires, including amplitudes and any correlations with the rest of the register. It does not copy either state. Three CNOTs can decompose SWAP, but the QPU system provides it as one primitive instruction. Applying SWAP twice restores the original ordering.

SWAP -I A B -O A B

The two output tokens are mandatory and must resolve to the same two wires, in the same order, as the inputs.

a	b	a'	b'
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	1

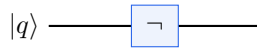
9 Derived logical gates

Derived gates are QPU instructions implemented as reversible target mutations. Inputs are preserved. If target t starts in zero, AND/OR/XOR write the familiar classical function. If it does not, the true behavior is the XOR embedding shown below.

One shape for every two-input gate

AND, NAND, OR, and XOR all follow $t' = t \oplus f(a, b)$. The two inputs a and b (**Control A** and **Control B** in the workbench) are read and left unchanged. The target t (**Target particle**) is the output workspace and is the only wire that changes. Only f differs between the four gates. To get the ordinary classical answer, start the target at 0 with SET Out Op.

9.1 Logical NOT (NOT)



$$t' = t \oplus 1.$$

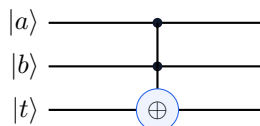
NOT is the derived logical spelling of a bit flip. In the current simulator it uses the same X matrix on the output target. The required input establishes the one-input instruction shape, but a distinct input is not copied into the output; use the same token on both sides.

NOT -I Target -O Target

t	t'
0	1
1	0

The current gate implementation mutates the output target. Use the same token for input and output; a distinct input token is not copied.

9.2 Reversible AND (AND)



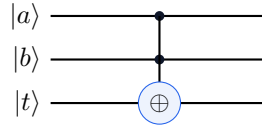
$$t' = t \oplus (a \wedge b).$$

AND flips the target exactly on the basis components where both controls are one. This is Toffoli-style behavior. It remains linear on superpositions and may entangle the target with its controls. Calling it AND is convenient only when the target is deliberately initialized.

SET Out Op
AND -I A B -0 Out

a	b	$t = 0$	$t' = a \wedge b$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1

9.3 Reversible NAND (NAND)



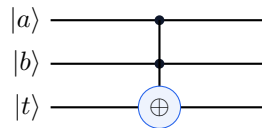
$$t' = t \oplus (a \wedge b) \oplus 1.$$

NAND first performs the all-controls-active mutation and then inverts the target. With target zero it produces the familiar NAND column. With target one it instead produces AND, illustrating why reversible target state must always be included in the full mathematical definition.

SET Out Op
NAND -I A B -0 Out

a	b	$t = 0$	$t' = \neg(a \wedge b)$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	0	0

9.4 Reversible OR (OR)



$$t' = t \oplus (a \vee b).$$

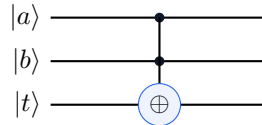
OR uses an “any control is one” predicate. If either or both controls are active, the target is flipped. This direct derived operation is equivalent on basis states to combining XOR and AND through $a \vee b = a \oplus b \oplus (a \wedge b)$.

SET Out Op
OR -I A B -0 Out

a	b	$t = 0$	$t' = a \vee b$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	0	1

For bits, $a \vee b = a \oplus b \oplus (a \wedge b)$.

9.5 Reversible XOR (XOR)



$$t' = t \oplus a \oplus b.$$

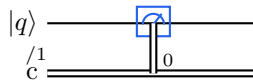
XOR flips the target when an odd number of controls is one. With two inputs, this is the ordinary exclusive-or predicate. The parity definition explains why the 11 row does not flip: two active controls have even parity.

```
SET Out Op
XOR -I A B -O Out
```

a	b	$t = 0$	$t' = a \oplus b$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	0	0

10 Measurement gate

10.1 Measurement (MEASURE)



```
MEASURE -I Q      # exactly one named wire
MEASURE           # every wire known at this point
```

MEASURE takes no -O. Repeated one-wire commands are useful when only declared returns should become classical. A bare command emits one measurement operation per currently allocated wire.

State	Outcome probabilities	Collapsed state
$ 0\rangle$	$P(0) = 1$	$ 0\rangle$
$ 1\rangle$	$P(1) = 1$	$ 1\rangle$
$\alpha 0\rangle + \beta 1\rangle$	$P(0) = \alpha ^2, P(1) = \beta ^2$	$ 0\rangle$ or $ 1\rangle$

Three ways to measure in the builder

The meter A step *in the circuit*, drawn on the qubit wire. A double stroke drops from that column to the **c** lane and numbers the bit that was read. **c** records the time of the measurement. The qubit wire stays a single line, and later gates on that particle still run. The meter is saved in the protocol as **MEASURE**.

Measure target Measures the selected **Target particle** once, right now. Nothing is added to the circuit.

Measure all Measures every wire not yet measured, right now.

MEASURE, like RESET, is not reversible: once a superposition collapses, the lost amplitudes cannot be recovered. Choose **Reset state** to rewind and sample again.

11 State, token, and cycle commands

11.1 SET — initialize or alias (lowered/executed)

```
SET Q Op
SET Q 1p
SET Q sp
SET Local $Parameter
```

SET **target value** requires a positional value. A constant schedules or emits initialization: zero, X from zero, or H from zero. A nonconstant value makes the target an alias of that source. A **state** parameter's constant SET is treated as a runtime default, allowing the UI or truth-table runner to supply its actual start state.

Syntax: exactly one target and one value are meaningful. SET has no **-I** or **-0** flags. It does not perform classical assignment at each simulation step; it establishes compilation-time preparation or reference identity. See the dedicated constants and aliases chapters before using SET on unknown quantum data.

11.2 CREATETOKEN — reserve wires (lowered/executed)

```
CREATETOKEN -I Sum Carry Workspace
```

Every **-I** token immediately claims a stable wire index. No **-0** is used. This is useful for declared outputs before their first gate.

CREATETOKEN does not initialize amplitudes by itself. Pair new workspace with SET before relying on its value. Listing outputs early also makes process structure easier to review:

```
CREATETOKEN -I Sum Carry
SET Sum Op
SET Carry Op
```

11.3 DELETETOKEN — end a token name (lowered/executed)

```
DELETETOKEN -I Workspace
```

The named token leaves scope. A later use is a compile error. The wire is returned to the allocator only when it is already known to be $|0\rangle$, which is the state after **SET wire Op** and before any gate targets it. Deletion does not emit RESET and does not erase an unknown or entangled state.

11.4 FREE — same lifetime rule (lowered/executed)

```
FREE -I ScratchA ScratchB
```

FREE uses the same rule as DELETETOKEN. Use **SET wire Op** when the intended behavior is zero preparation. A wire that is not known $|0\rangle$ keeps its qubit index, and only the name dies.

11.5 INCREASECYCLE — cycle boundary (lowered/executed)

INCREASECYCLE

The compiler flushes pending zero-initializations into an internal batched **RESET**, increments the cycle counter, and prepares workspace for following operations. Cycle suffixes on tokens do not create separate visual wires.

Cycles organize protocol stages. They are not wall-clock timing guarantees and do not cause gates to execute concurrently. The final circuit is still an ordered gate sequence. Each emitted gate records the timeline cycle, and the canvas draws a dashed vertical line across every wire at **INCREASECYCLE**, labelled **IC** with the new cycle number (**IC1**, **IC2**, ...). The line turns red while that step is active. Inside a collapsed recursive call these markers are not drawn; the teal depth badge replaces them. An integer suffix such as **Q:1** must match this process's cycle. A child process starts again at cycle 0, so its own **:0** labels stay valid after the parent advances. Non-numeric suffixes stay labels. The suffix does not allocate another wire.

11.6 Cycle-based state evolution

Cycle does *not* mean loop. Cycle does not mean parallel execution. Cycle does not mean wall-clock time. Cycle means the next logical stage of the same finite circuit:

Cycle 0 → **INCREASECYCLE** → Cycle 1 → **INCREASECYCLE** → Cycle 2

A classical sequential rule such as $Q_{\text{next}} = Q \oplus T$ is shown by *unrolling*:

```
CNOT -I T -0 Q
INCREASECYCLE
CNOT -I T -0 Q
INCREASECYCLE
CNOT -I T -0 Q
```

Conceptually $Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3$. There is no backward execution edge. The Interactive Workbench button **Add cycle boundary** inserts the same marker visually. The starter circuit *T state transition across cycles* demonstrates this pattern.

11.7 JOIN — name an ordered register (lowered/executed)

JOIN -I A B -0 AB

JOIN gives the ordered wires A then B the name AB. It is not a gate and allocates no amplitudes: the simulator state is already one vector in $(\mathbb{C}^2)^{\otimes n}$. At least two inputs and exactly one output are required. A one-qubit gate on a multi-wire register is an error; name one wire or **SPLIT** it first.

11.8 SPLIT — name a subsystem (lowered/executed)

SPLIT AB A 2

The third argument is a power-of-two dimension, the same convention as **_dim**. 2 peels off one qubit from the front of AB and names it A. The remaining wires stay on AB. **SPLIT** does not copy amplitudes. An entangled register still has no separate pure state for A; the new name is a view of those wires inside the global vector.

12 Reset: an operation that is not a gate

The word *reset* appears in two unrelated places: the compiler's internal **RESET** operation, and three buttons in the Circuit builder. Neither is an ordinary quantum gate.

12.1 Reversible gates versus RESET

Every gate in the palette except the meter is *reversible*: its effect can always be undone. X, Y, Z, H, CNOT, CCNOT, CZ, CY, SWAP, NOT, AND, NAND, OR, and XOR undo themselves when applied a second time. S, T, and PHASE are undone by the same phase with the opposite sign. Mathematically each is a *unitary* matrix U , and $U^\dagger$ undoes it. RESET is different. It forces one wire to $|0\rangle$ whatever it held before:

$$|0\rangle \mapsto |0\rangle, \quad |1\rangle \mapsto |0\rangle, \quad \alpha|0\rangle + \beta|1\rangle \mapsto |0\rangle.$$

Two different inputs, $|0\rangle$ and $|1\rangle$, give the same output, so nothing can tell afterwards which one was there. The information has been erased, and no matrix U can undo that. RESET is therefore not unitary, just like MEASURE.

	Reversible gates (X, H, MEASURE CNOT, AND, ...)		RESET
Unitary?	Yes	No	No
Can be undone?	Yes, by the inverse gate	No	No
Result depends on chance?	No	Yes	No
Written in source?	Yes	Yes	No — write SET wire 0p
Shown on canvas?	Yes. Inverse copies add †, a blue outline, and purple when active.	Yes, as the meter on the qubit, with a double stroke down to c	No (hidden)

12.2 Where RESET comes from

You never type RESET. The compiler inserts it when a wire must start at zero:

- SET Out 0p on a workspace wire. (SET Out 1p and SET Out sp are instead prepared by applying X or H to a wire that starts at 0.)
- INCREASECYCLE, which zeroes pending workspace before the next stage.
- RUNCHILD and CALL, which zero each parent output wire once before expanding the child.

RESET steps are hidden from the canvas and filtered from the **Execution log** so that the visible circuit shows only the gates you wrote. They still run during simulation.

Reset a wire only when it is not entangled

In this simulator, RESET keeps the part of the state in which the wire is already 0 and rescales it; only if that part is empty does it move the $|1\rangle$ amplitudes to $|0\rangle$. On a fresh wire this is exactly “prepare $|0\rangle$ ”. On a wire entangled with others it also changes the other wires. For example, resetting one half of a Bell pair forces the other half to 0 as well. In multi-stage circuits, *uncompute* temporary wires back to 0 with reversible gates instead of relying on a reset.

12.3 Reset buttons compared

None of these buttons inserts a RESET operation into your circuit. They only change what the page is showing and simulating.

Kept or cleared?	Reset state	Clear circuit	Reset site	Notes
Gates on the canvas	kept	cleared	cleared	
Number of wires	kept	kept	back to 3	
Start states (<i>name</i> start)	kept	kept	all 0p	
Protocol editor text	kept	rewritten	default	Clear circuit writes an empty canvas protocol into the editor.
Compiled PARAMS / RETURNVALS	kept	cleared	cleared	
Quantum state and measurements	rewound	rewound	rewound	Back to the start states; log restarts.
Workbench selections	kept	kept	defaults	Gate H, target q0, controls q1 and q2, phase 90°.
Custom gates and catalog	kept	kept	kept	Nothing removes these except Remove or closing the browser session.

Which one do I want?

Reset state “Run the same experiment again.” Use it between repeated runs of a circuit that measures, to collect several samples.

Clear circuit “Keep my wires, start a new circuit.”

Reset site “Put everything back the way it was when I first opened the page.” The **Reset site** entry in the navigation menu and **Reset site completely** on the **More** page do the same thing.

13 Process composition and return values

Main process and child process in plain words

A **process** is a named list of gates with inputs (PARAMS) and outputs (RETURNVALS). The **main process** is the one you compile and run: the name after MAIN-PROCESS. A **child process** is another saved process that the main process borrows with RUNCHILD, the way a program calls a helper function.

The difference from a function call is that nothing is called at run time. The compiler copies the child’s gates into the main circuit, on the wires you pass: `-I` wires feed the child’s inputs in PARAMS order, and `-O` wires receive its outputs in RETURNVALS order. The canvas therefore shows every gate, including the ones that came from children.

“Main” and “child” are roles in one compile, not kinds of file. **SingleBitFullAdder** is the main process when you run it on its own and a child when **TwoBitFullAdder** runs it twice. A custom gate is the canvas version of the same idea: a saved process placed on wires you choose.

Browse a process and its children

In the Circuit builder, press the **TwoBitFullAdder** protocol button. The Interactive workbench gains a **Compiled protocol** card showing the row of its truth table that matches the current start states. Open **Learn more** on that card for the full 32-row table (grouped by the leading input bits), its notes, and a **SingleBitFullAdder** entry you can expand in place to read the child.

13.1 MAIN-PROCESS (lowered/executed)

MAIN-PROCESS FullAdder

This marks the process body and supplies its catalog/library name. Compilation already begins from the named process, so the row itself emits no gate. The name is the first positional word after MAIN-PROCESS; avoid spaces inside it. This name connects a protocol to catalog entries, child calls, downloads, and companion truth-table metadata. If no marker exists, the system uses **InlineProcess**, but named processes are strongly recommended.

13.2 DECLARECHILD (lowered/executed)

```
DECLARECHILD SingleBitFullAdder
```

The named child is bound from the compilation `librarySources` catalog. An unknown name fails on this line. The row itself emits no gate; later `RUNCHILD` or `CALL` expansions use the bound process body. A successful declaration therefore guarantees that the named child exists in the catalog supplied to this compile, and that later calls of that name expand the bound body rather than a later alias.

13.3 RUNCHILD — expand a child process (lowered/executed)

```
RUNCHILD SingleBitFullAdder \
  -I A B Cin \
  -O Carry Sum
```

The first positional argument is the child process name. Inputs bind by the child’s `PARAMS` order: `A` maps to the first parameter, `B` to the second, and `Cin` to the third in this example. Outputs bind by the child’s `RETURNVALS` order. Before expansion, each bound parent output is zero-prepared once. Child-local names receive a unique scope. `RUNCHILD` is compile-time composition, not a dynamic subroutine call in a running quantum machine. The child body is recursively expanded into the parent’s flat gate sequence. This allows the renderer and simulator to inspect every resulting gate.

13.4 CALL — alternate child invocation (lowered/executed)

```
CALL SingleBitFullAdder -I A B Cin -O Carry Sum
```

`CALL` currently follows the same child lookup, parameter binding, output preparation, and expansion path as `RUNCHILD`. `RUNCHILD` is used by the bundled multi-bit protocols and makes compile-time expansion explicit; `CALL` is retained as an equivalent system spelling. Both fail when the named child is absent from the supplied process library.

13.5 RETURNVALS (lowered/executed)

```
RETURNVALS Carry Sum
```

Positional tokens define the process’s ordered public outputs. They determine child output binding, truth-table output columns, and the UI’s preferred logical width. Do not use `-I` or `-O` here: flags would be retained as positional return tokens.

`RETURNVALS` does not measure automatically. Add `MEASURE` instructions when classical outcomes are required. Returning an unmeasured wire preserves it as part of the projected quantum output shown by the system.

13.6 ACCEPTVALS (lowered/executed)

```
CALL Child -I A B
ACCEPTVALS LocalCarry LocalSum
```

Each positional local token aliases the corresponding value from the most recent child return list. Direct `-O` binding on `RUNCHILD`/`CALL` is clearer when output names are known.

Ordering is again essential: the first `ACCEPTVALS` name receives the first returned reference, and so on. The instruction refers to the most recently completed child expansion in compiler state; inserting another child call changes which return list is accepted.

13.7 Nested full-adder example

```

# child.qpucir
PARAMS: A:state B:state Cin:state
MAIN-PROCESS ChildAdder
CREATETOKEN -I Carry Sum
SET Carry Op
SET Sum Op
XOR -I A B -O Sum
CNOT -I Cin -O Sum
CCNOT -I A B -O Carry
CCNOT -I A Cin -O Carry
CCNOT -I B Cin -O Carry
RETURNVALS Carry Sum

# parent.qpucir
PARAMS: A:state B:state Cin:state
MAIN-PROCESS Parent
CREATETOKEN -I Cout Result
RUNCHILD ChildAdder -I A B Cin -O Cout Result
MEASURE -I Cout
MEASURE -I Result
RETURNVALS Cout Result

```

For basis inputs, the full-adder equations are

$$\text{Sum} = A \oplus B \oplus C_{\text{in}}, \quad C_{\text{out}} = (A \wedge B) \oplus (A \wedge C_{\text{in}}) \oplus (B \wedge C_{\text{in}}).$$

A	B	C _{in}	C _{out}	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

13.8 Bounded child-process recursion (lowered/executed)

The QPU protocol does not support runtime loops or recursive execution of the compilation root. A child marked REC may recursively expand itself during compilation only. The parent supplies a finite -DEPTH; each RECUR creates another compile-time frame with DEPTH - 1. EXIT WHEN ... is the compile-time base case. After expansion the simulator still receives a flat ordered gate list.

```

# recursive-h.qpucir
PARAMS: Q:state
MAIN-PROCESS RecursiveH
REC MAXDEPTH 16
EXIT WHEN DEPTH == 0
H -I Q -O Q
INCREASECYCLE
RECUR -I Q
RETURNVALS Q

```

```

# parent
PARAMS: Q:state
MAIN-PROCESS Parent
DECLARECHILD RecursiveH

```

```
RUNCHILD RecursiveH -DEPTH 4 -I Q
RETURNVALS Q
```

Compiler structure:

Parent $\rightarrow$ RecursiveH DEPTH 4 $\rightarrow$ DEPTH 3 $\rightarrow$ DEPTH 2 $\rightarrow$ DEPTH 1 $\rightarrow$ stop

Emitted circuit (no recursion metadata remains):

```
H
INCREASECYCLE
H
INCREASECYCLE
H
INCREASECYCLE
H
INCREASECYCLE
```

Rules (summary):

- Root $\rightarrow$ root is forbidden.
- Child $\rightarrow$ same child is allowed when bounded.
- Mutual recursion ($A \rightarrow B \rightarrow A$) is rejected.
- RECUR requires REC or TREC; the initial RUNCHILD requires -DEPTH.
- DEPTH, LEVEL, and ROOTDEPTH are read-only.
- Absolute compiler limit: 64; optional REC/TREC MAXDEPTH n is lower.
- In-place -I (same wire returned) is not zero-prepared between recursive frames.
- **TCO (F#-style)**: if every RECUR is in tail position, REC automatically expands iteratively in one compile-time frame (looping rewind). Explicit TREC asserts the same form and errors if post-RECUR work exists. Non-tail REC keeps stacked frames.

Keep these concepts distinct:

Construct	Meaning
RUNCHILD	Compose another process
RECUR	Expand this child again with DEPTH-1
REC	Allow recursion; auto-TCO when tail
TREC	Require tail form; always TCO
INCREASECYCLE	Next logical circuit stage (not a loop)
WHEN	Compile-time condition on DEPTH/LEVEL/ROOTDEPTH
-IF	Runtime classical feed-forward after MEASURE
IF/ELSE/ENDIF	Block form of -IF with an alternative branch

The existing bit adders remain the teaching example of explicit hierarchical composition. Recursion is a circuit-*construction* abstraction for repeated stages; it never introduces a backward execution edge.

13.8.1 Recursion on the circuit canvas

The compiled gate list holds every unrolled frame, but the canvas draws each recursive call once:

- A single-gate body (such as **RecursiveH**) keeps its own symbol. A body with several gates collapses to one **REC** box that spans every wire the body touches. Above the box, a caption gives the child name and the depth range, for example **RecursiveReversibleEcho**, D5 $\rightarrow$ D1.
- A teal **D n** badge above the gate shows the current DEPTH. It reads the root depth before the call, counts down (D5, D4, ...) as **Step gate** walks through the frames, and hides once the call has finished.
- The REC box follows the usual colours: red while a forward gate of the frame is running, purple while an inverse (dg/inv) gate is running.
- The INCREASECYCLE markers inside the expansion are not drawn on the canvas; the badge replaces them.

While the playhead is inside a recursive call, the Interactive workbench shows a **Recursive frame** card. It draws the level being executed with the same gate symbols as the canvas: the heading gives **DEPTH** d · **LEVEL** ℓ · **TCO** (or **STACK**), the forward gates sit above a **forward** $\rightarrow$ rule and the adjoint gates above a $\leftarrow$ **inverse** rule, the frame's **INCREASECYCLE** is a dashed line labelled ICn , and the frame ends in **RECUR** $\downarrow D(d-1)$ (**DO exit** on the last frame). The current gate is highlighted. When the playhead is elsewhere the card is collapsed as *Expand ... first frame*. The downward arrow is deliberate: the next frame is the next block of the finite circuit, not a jump back to an earlier column.

13.9 Structured classical branches: IF / ELSE / ENDIF (lowered/executed)

A block conditions every gate inside it on one measured bit, with an optional alternative:

```
MEASURE -I A
IF A=1
X -I C -O C
ELSE
Z -I C -O C
ENDIF
```

The compiler removes IF, ELSE, and ENDIF and emits ordinary gates. Each gate in the IF part carries the condition $A=1$; each gate in the ELSE part carries the opposite value, $A=0$. The example above compiles exactly as

```
MEASURE -I A
X -I C -O C -IF A=1
Z -I C -O C -IF A=0
```

plus canvas metadata that records which gates came from the IF part and which from the ELSE part.

Rules:

- The header is IF Token=0 or IF Token=1, with no spaces around =. The token may be a parameter or any name that is in scope.
- ELSE is optional and takes no arguments. It flips the value for the gates after it. ENDIF is required and takes no arguments.
- ELSE or ENDIF without a matching IF, or an IF left open at the end of the process, is a compile error.
- The token must be measured before the first conditioned gate runs; otherwise the simulator stops with “requires qn to be measured first”.
- Only the innermost open block controls a gate. Blocks may be written inside each other, but the inner condition *replaces* the outer one; it is not combined with it. A gate's own -IF inside a block is also replaced by the block's condition.
- Both branches remain in the compiled forward sequence. At run time the gate whose condition fails is skipped (the run log says “skipped because classical condition was false”); nothing is removed and nothing forks.

13.9.1 IF / ELSE on the circuit canvas

The quantum wire stays a single straight line. The branch gates are drawn one after another on that wire, never as a fork:

- A yellow double line drops from each conditioned gate to the classical **c** lane. It passes behind any gate it crosses.
- Under the **c** lane each branch is labelled with its keyword and test, for example **IF A=1** and **ELSE A=0**. Compiled protocols use the parameter name; canvas circuits use qn .
- Before the bit is measured both branches are drawn normally. After the measurement the branch that runs shows ✓ **taken**; the other shows ∅ **skipped**, and its gate, feed line, and label fade and turn dashed. If the measured value changes, the two swap.

13.10 Worked example: recursion, measurement, and IF / ELSE together

The bundled **RecursiveReversibleEchoHarness** protocol combines both features. Its child applies a forward block and then the adjoints of that block in reverse order, so each frame returns the wires to where they started:

```
# recursive-reversible-echo.qpucir (abridged)
PARAMS: A:state B:state C:state
MAIN-PROCESS RecursiveReversibleEcho
TREC MAXDEPTH 8
EXIT WHEN DEPTH == 0
H -I A -O A
RX=pi/3 -I B -O B
RY=pi/5 -I C -O C
CNOT -I A -O B
CPHASE=pi/7 -I B -O C
T -I A -O A
AND -I A B -O C
SWAP -I B C -O B C
SWAPdg -I B C -O B C
ANDinv -I A B -O C
Tdg -I A -O A
invCPHASE=pi/7 -I B -O C
CNOTdg -I A -O B
RYinv=pi/5 -I C -O C
dgRX=pi/3 -I B -O B
Hinv -I A -O A
INCREASECYCLE
RECUR -I A B C
RETURNVALS A B C
```

The harness prepares $|101\rangle$, runs the child with `-DEPTH 5`, and then branches on the measured value of A:

```
X -I A -O A
X -I C -O C
RUNCHILD RecursiveReversibleEcho -DEPTH 5 -I A B C -O A B C
ACCEPTVALS A B C
MEASURE -I A
IF A=1
X -I C -O C
ELSE
Z -I C -O C
ENDIF
MEASURE -I B
MEASURE -I C
```

Because every frame undoes itself, A is still 1 after the echo, so the IF branch runs: C flips to 0 and the final reading is A=1, B=0, C=0. On the canvas the call is one REC box whose badge counts $D5 \rightarrow D1$; the X on C shows ✓ **taken** and the Z shows ∅ **skipped**.

14 Custom gates: packaging a circuit as one block

A custom gate is a whole QPU process saved under a short name so that it can be placed like any built-in block. It adds no new physics: when it runs, the simulator replays the process's own gates, one by one, on the wires you chose. Understanding a custom gate therefore means understanding the smaller gates inside it.

14.1 Register one in four steps

1. Paste the process below into the protocol editor in **Compile text into a circuit** and choose **Compile protocol** to confirm it has no errors.
2. In **Register custom gates from QPU processes**, type NOR in **Gate id (palette label)** and leave **Source process** on *Current protocol editor*.
3. Choose **Register custom gate**. A **NOR** block appears under **Custom gates** in the palette and in the workbench **Gate** list.

4. Place it with **Control A**, **Control B**, and **Target particle**, exactly like AND.

Complete runnable protocol

PARAMS: A:state B:state

MAIN-PROCESS NorGate

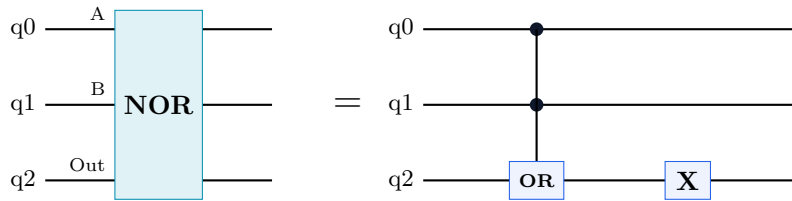
CREATETOKEN -I Out

OR -I A B -O Out

X -I Out -O Out

RETURNVALS Out

14.2 How the process maps onto wires



In the process	Becomes, when the block is placed
1st PARAMS entry	Control A (the lowest free wire when dropped on the canvas).
2nd PARAMS entry	Control B . Further entries take the next free wires.
1st RETURNVALS entry	Target particle : the only wire the block should change.
Any other token	A fresh extra workspace wire that the block allocates for itself.

The palette gives every custom block exactly one target wire, so register processes with a single **RETURNVALS** output. A process with two outputs, such as a half-adder, compiles on its own but cannot be placed as a block; use it through **RUNCHILD** instead.

14.3 Reasoning about reversibility

A sequence of reversible gates is reversible: undo the last gate first, then the one before it, and so on.

$$(U_k \cdots U_2 U_1)^{-1} = U_1^{-1} U_2^{-1} \cdots U_k^{-1}.$$

NorGate is OR followed by X. Each gate only XORs a value into Out, so the two gates commute and NOR undoes itself: placing the block twice in a row returns every wire to where it started. Two self-inverse gates in sequence are not self-inverse in general; the inverse of $U_2 U_1$ is $U_1 U_2$. Written as a rule, the block performs

$$\text{Out}' = \text{Out} \oplus \neg(A \vee B),$$

the same “target XOR function” shape as AND. That shape is always reversible, because applying it again XORs the same value back out.

Three things that make a custom gate irreversible

MEASURE inside the process The block collapses its wires every time it runs.

SET on the output SET Out 0p makes the compiler insert a hidden RESET, so the block *overwrites* the target ($\text{Out}' = f(A, B)$) instead of XORing into it. Whatever the target held before is erased. NorGate deliberately has no SET: a freshly compiled wire already starts at 0, so it passes the same truth table.

Temporary wires left dirty Helper tokens become hidden workspace wires. If a helper is not returned to 0 (uncomputed), it stays entangled with the inputs, and the visible wires alone no longer behave reversibly.

A quick test you can run in the builder

Place your block twice on the same wires and choose **Run all**. For a reversible out-of-place gate, every wire ends exactly as it started. Try it with the target's *name* start set to **0p** and then **1p**. If the target does not return, look for one of the three problems above.

14.4 Rules to remember

- The gate id must start with a letter, use only letters, digits, and underscores, and cannot match a built-in gate such as **AND**.
- Custom gates are saved for the current browser session only, not in the bundled catalog. Keep the `.qpucir` file to recreate one later.
- A custom gate can wrap a process that itself uses **RUNCHILD**; the child processes are captured when you register it.

15 Process control and checkpoints

15.1 COMPILEPROCESS (lowered/executed)

COMPILEPROCESS ChildName

COMPILEPROCESS looks up the named process and compiles it on a scratch state. Its gates are not inserted into the current circuit. Use **RUNCHILD** or **CALL** when the child should actually run. An unknown name is an error. A process that asks to compile itself is verified once and does not recurse.

15.2 MASTERVAL (lowered/executed)

MASTERVAL Result

When the process has no **RETURNVALS**, **MASTERVAL** names are the ordered public outputs. When **RETURNVALS** is present, every **MASTERVAL** name must already appear there, and the truth-table columns stay the **RETURNVALS** list.

15.3 Save-state checkpoint instruction (lowered/executed)

SAVE_STATE checkpointA

SAVE_STATE copies the current amplitude vector and measurement record into a classical store under that name. It is not a unitary and does not clone the quantum state inside the register.

15.4 Load-state checkpoint instruction (lowered/executed)

LOAD_STATE checkpointA

LOAD_STATE replaces the active state vector and measurement record with the named checkpoint. A missing name is an error. Later gates continue from the restored amplitudes.

On the canvas neither instruction is a gate box. Each is a dashed vertical line across every wire and the **c** lane, labelled **S** (save) or **L** (load) with the checkpoint name above the letter. Gates between a matching S and L still run, but their effect is discarded when the checkpoint is loaded.

16 Truth-table companion syntax

Truth tables are metadata paired to a process; they are not `.qpucir` commands. A protocol describes *how* a circuit works; its companion `.qpuio` describes selected input/output expectations.

16.1 Text `.qpuio` structure

A text table may use spaces or commas:

```
MAIN-PROCESS: HalfAdder
INPUTS: A B
OUTPUTS: Carry Sum
# A B Carry Sum
0 0p 0p 0p 0p
1 0p 1p 0p 1p
2 1p 0p 0p 1p
3 1p 1p 1p 0p
```

16.2 MAIN-PROCESS: table header

The colon is required in the table format:

```
MAIN-PROCESS: HalfAdder
```

This name associates metadata with its protocol. Notice the deliberate syntax difference: protocol source writes `MAIN-PROCESS HalfAdder`, while the truth-table header writes `MAIN-PROCESS: HalfAdder`.

16.3 INPUTS: and OUTPUTS: column groups

```
INPUTS: A B
OUTPUTS: Carry Sum
```

These lines divide the columns into values supplied before simulation and values checked after simulation. Their order must match the later column header. INPUTS should correspond to state-typed PARAMS; OUTPUTS should correspond to RETURNVALS. OUTPUTS must contain at least one name.

Explicit groups remove ambiguity. If they are absent, the loader can derive the split only when a matching protocol is available and the combined PARAMS and RETURNVALS widths match the table header.

16.4 Column header row

```
# A B Carry Sum
```

The first nonmetadata row beginning with a hash and whitespace or comma is the ordered column header. Here A and B are inputs, followed by Carry and Sum outputs. Unlike an ordinary protocol hash comment, this row has structural meaning inside a `.qpuio` file.

16.5 Indexed data rows

```
0 0p 0p 0p 0p
1 0p 1p 0p 1p
```

The first cell is a decimal row index. Indices begin at zero and increase by one without gaps. Remaining cells align left-to-right with the column header. Every row must have exactly the expected width, and every state cell must be `0p`, `1p`, or `sp`.

For n binary input columns, a complete table contains 2^n rows. The usual ordering interprets input columns most-significant first: for two inputs the sequence is 00, 01, 10, 11. The loader checks sequential row indices, while semantic validation also rejects duplicate input patterns.

16.6 CSV spelling

Commas may replace whitespace consistently:

```
MAIN-PROCESS: HalfAdder
INPUTS: A,B
OUTPUTS: Carry,Sum
#,A,B,Carry,Sum
0,0p,0p,0p,0p
```

Spaces around commas are trimmed. Do not mix extra prose into a data row; trailing hash comments are allowed only where the table loader can strip them after the data content.

16.7 Partial truth tables

A partial table lists selected cases rather than all 2^n combinations. It must contain at least one row, no duplicate input pattern, and no more than 2^n rows. Partial tables are useful for state-machine transitions or focused regressions, but passing one proves behavior only for the listed cases.

Input count n	Complete row count 2^n	Example
0	1	constant-output process
1	2	0, 1
2	4	00, 01, 10, 11
3	8	000 through 111
4	16	0000 through 1111

16.8 JSON .qpuio envelope

The equivalent machine-oriented JSON form is:

```
{
  "format": "qpuio",
  "version": 1,
  "processName": "HalfAdder",
  "inputColumns": ["A", "B"],
  "outputColumns": ["Carry", "Sum"],
  "rows": [
    ["0p", "0p", "0p", "0p"],
    ["0p", "1p", "0p", "1p"],
    ["1p", "0p", "0p", "1p"],
    ["1p", "1p", "1p", "0p"]
  ]
}
```

The `format` value must be `qpuio` and `version` must be 1. Column names must be nonempty strings, `outputColumns` must not be empty, and every nested row array must have exactly

$$\text{inputColumns.length} + \text{outputColumns.length}$$

valid cells.

16.9 How a table test runs

For each listed row, the runner:

1. maps input cells to state PARAMS by column name and order;
2. creates the circuit's start-state vector;
3. executes every compiled gate, including child expansions;
4. measures the declared RETURNVALS;
5. compares only those measured outputs with expected output cells.

An expected output **sp** accepts either measured bit. An input **sp** initializes an equal plus-state superposition. Because one probabilistic sample cannot establish a full distribution, tables are strongest for deterministic basis-state behavior.

16.10 Protected bundled tables

The bundled SingleBitFullAdder, TwoBitFullAdder, FourBitFullAdder, and PhaseDemo tables are canonical site assets. Do not rewrite them to make an experiment pass. Create a new uploaded or corrected process/table pair when testing different expectations. This keeps examples reproducible and prevents a protocol correction from silently changing its specification.

17 Compiler output and lowering details

Compilation returns:

```
CompileResult {
  gates: CircuitGate[]
  qubitCount: number
  logicalQubitCount: number
  parsed: ParsedCommand[]
  log: string[]
  tokenMap: Record<string, number>
  processParams: ProcessParam[]
  returnValues: ReturnValue[]
}
```

Each accepted instruction entry records **op**, original **raw** text, **inputs**, **outputs**, positional **args**, optional normalized radian **phase**, **reverse**, and **noParameterSubstitution**. Each emitted gate records **id**, **type**, ordered **step**, **targets**, **controls**, optional **phase**, and source text.

After child expansion, unused symbolic holes are compacted to dense qubit indices. Internal RESET gates count for simulation but are filtered from the visible circuit. UI logical width prefers RETURNVALS count, then exposed PARAMS count, then physical simulator width.

18 Error conditions and practical rules

1. Unknown opcodes fail with **Unknown command**. RESET is not a source opcode.
2. Every QPU gate except MEASURE requires **-I** and **-O**, then exact registry arity is checked.
3. PHASE requires a valid nonzero-denominator angle after **=**.
4. SWAP requires two matching input/output wires.
5. RUNCHILD/CALL fails if the named process is absent from the library. Unknown DECLARECHILD names fail on the declaration line.
6. SET fails when its positional value is missing.
7. RETURNVALS order is semantically significant for child bindings and truth-table columns.
8. Use the same input/output token for single-qubit gates.
9. Initialize derived-gate targets deliberately; their operation is XOR into the existing target, not destructive assignment.
10. FREE and DELETETOKEN end a name. They reuse a wire only from a known $|0\rangle$.

19 Quick-reference grammar

This is a practical description of the accepted surface language rather than a separate formal language specification:

```

protocol      := [params-line] logical-line*
params-line   := "PARAMS:" (name ":" type)*
logical-line  := command | comment | blank

command       := opcode argument*
opcode        := supported-command
               | primitive-gate
               | primitive-gate ("dg" | "inv")
               | ("dg" | "inv") primitive-gate
               | "PHASE=" angle
               | "PHASE" ("dg" | "inv") "=" angle
               | ("dg" | "inv") "PHASE=" angle

gate-row      := opcode "-I" input+ "-O" output+ ["-$R"]
               ["-IF" name "=" ("0" | "1")]
measure-row   := "MEASURE" ["-I" input]
set-row       := "SET" target value
process-row   := "MAIN-PROCESS" name
child-row     := ("RUNCHILD" | "CALL") name ["-DEPTH" integer]
               ["-I" argument*] ["-O" argument*]
rec-row       := ("REC" | "TREC") ["MAXDEPTH" integer]
exit-row      := "EXIT" "WHEN" ("DEPTH" | "LEVEL" | "ROOTDEPTH")
               ("==" | "!=" | "<" | "<=" | ">" | ">=") integer
recur-row     := "RECUR" "-I" argument+ ["-O" argument*]
if-block      := "IF" name "=" ("0" | "1")
               logical-line*
               ["ELSE" logical-line*]
               "ENDIF"
returns-row   := "RETURNVALS" token*
accept-row    := "ACCEPTVALS" token*

token         := ["$"] name [":" suffix]
constant      := ("0p" | "1p" | "sp") ["_dim" integer]
angle         := rational
               | rational "d"
               | [sign] [integer ["*"]] "pi" ["/" integer]

```

Syntax detail: PHASE= pi/2 is joined into one opcode/value pair, but PHASE = pi/2 is not. Operations and flags are case-insensitive; names, types, and positional values are otherwise retained.

Block and recursion detail: -DEPTH is required when the child declares REC or TREC. RECUR, REC, TREC, and EXIT WHEN are only meaningful inside a child process. An if-block lowers each inner gate to a -IF condition; see Structured classical branches.

20 Glossary

Term	Meaning
Amplitude	A complex coefficient of a basis state. Its squared magnitude contributes to measurement probability.
Ancilla	An auxiliary qubit or workspace wire, usually prepared in a known state, used to hold intermediate information in a reversible computation.
Basis state	A state used as a coordinate direction. The computational basis for one qubit is $ 0\rangle, 1\rangle$.

Term	Meaning
Bloch sphere	A geometric representation of pure one-qubit states after ignoring global phase. North/south are $ 0\rangle/ 1\rangle$; longitude represents relative phase.
Boolean algebra	The mathematics of 0/1 values and operations such as NOT, AND, OR, and XOR.
Child process	A separately named protocol expanded into a parent through RUNCHILD or CALL.
Combinational logic	A classical circuit whose output depends only on current inputs, not stored history.
Compile-time recursion	Bounded self-expansion of a child process marked REC or TREC. RECUR expands another frame with DEPTH-1; EXIT WHEN ends a frame. Tail-position REC auto-converts to TCO (iterative one-frame expansion, F#-style); TREC requires that form. No runtime loop or recursive QPU execution is created.
IF / ELSE / ENDIF	Structured classical feed-forward sugar. Compiles to flat gates carrying -IF conditions (ELSE gates get the opposite value) and branch metadata for the canvas labels. Both branches stay in the circuit; the one whose bit does not match is skipped and drawn faded as “skipped”. Never forks the quantum wire.
Control	A wire whose basis condition decides whether a controlled operation acts on its target. The control itself is preserved by the standard controlled gates.
DEPTH	Read-only remaining recursive expansions for the current child frame. Supplied by RUNCHILD -DEPTH and decremented by RECUR.
D_n badge	Teal label above a recursive call on the canvas showing the current DEPTH; counts down while stepping and hides when the call ends.
Entanglement	A joint multi-qubit state that cannot be written as a tensor product of independent qubit states.
Fan-out	Sending one classical value to multiple destinations. Unknown quantum states cannot be universally copied; CNOT copies only known basis information onto a zero target and otherwise may entangle.
Gate	A state transformation. Closed-system quantum gates are unitary; measurement and reset have different semantics.
Garbage	Reversible-workspace information no longer wanted as output but still present because it cannot simply be erased.
Global phase	A unit complex multiplier applied to the whole state. It does not change physical measurement predictions.
Ket	Dirac notation $ \psi\rangle$ for a state column vector.
Measurement	Conversion of quantum amplitudes into a sampled classical result, with state collapse and renormalization.
Phase	The angular part of a complex amplitude. Relative phase controls interference even when immediate basis probabilities are unchanged.
Process	A named QPU protocol body with ordered PARAMS and RETURNVALS.
Protocol	Plain-text QPU instructions describing preparation, gates, composition, measurement, and returned references.
Qubit	A normalized two-component complex state $\alpha 0\rangle + \beta 1\rangle$, not merely an unknown classical bit.
Register	One or more logical wires considered together. Current JOIN/SPLIT syntax is accepted but does not yet lower to register operations.
Relative phase	The phase difference between amplitudes. Unlike global phase, it can change later interference and measurement probabilities.
Superposition	A coherent linear combination of basis states. <code>sp</code> prepares the equal plus superposition.
Target	The wire whose amplitudes a gate mutates; normally the first -0 reference.
Tensor product	The operation $\otimes$ that combines state spaces. Two one-qubit vectors produce a four-amplitude two-qubit vector.
Token	A source-level logical name resolved to a qubit wire or alias during compilation.
Truth table	Listed input/output expectations for measured RETURNVALS. It describes basis behavior but not arbitrary phase or entanglement.

Term	Meaning
Uncomputation	Running suitable inverse gates in reverse order to return temporary workspace to a known state without irreversible erasure.
Unitary	A matrix satisfying $U^\dagger U = I$, preserving normalization and having a reversible inverse.

21 Operation checklist

Item	Covered in	Item	Covered in
PARAMS:	Lexical syntax	MAIN-PROCESS	Process composition
X,Y,Z,H	Single-qubit gates	S,T,PHASE,RX,RY,RZ	Single-qubit gates
CNOT,CCNOT	Controlled gates	CZ,CY,CPHASE,SWAP	Controlled gates
NOT,AND,NAND	Derived gates	OR,XOR	Derived gates
MEASURE	Measurement	SET	State/token commands
CREATETOKEN	State/token commands	DELETETOKEN,FREE	State/token commands
INCREASECYCLE	State/token commands	JOIN,SPLIT	State/token commands
DECLARECHILD	Process composition	RUNCHILD,CALL	Process composition
REC,TREC,RECUR,EXIT	Process composition	-DEPTH,WHEN	Process composition
IF,ELSE,ENDIF	Classical feed-forward	Token=0 1	Process composition
RETURNVALS	Process composition	ACCEPTVALS	Process composition
COMPILEPROCESS	Compatibility	MASTerval	Compatibility
SAVE_STATE	Compatibility	LOAD_STATE	Compatibility
-I,-O,-\$R	Flags	dg,inv	Single-qubit/inverse syntax
comments,\	Lexical syntax	.qpuio	Truth-table syntax